

Track 48: CNC Programming

June 2026

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Tier 3 — Specialized | Departments: Mechanical · Mechatronics

Skill path: CNC Basics → G-Code/M-Code → CAM → Toolpaths → 2.5D/3D Milling → Lathe → Simulation → Production

Related: [Track 52 — SolidWorks](#) · [Track 37 — CAD/CAM](#) · [Track 25 — PLC](#)

Track Overview

Who this is for: Mechanical and Mechatronics students targeting CNC operator/programmer and manufacturing engineer roles.

Prerequisites: Basic machining knowledge; CAD experience ([Track 52](#)) recommended.

Tools: CNC simulators (CNCSimulator/Fusion 360 CAM), G-code editors, 3-axis mill/lathe sim, Mastercam/Fusion CAM.

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	CNC awareness	G-code programs + simulation
1 Month	80–100	Junior CNC programmer	3 machined parts (sim)
3 Months	250–300	Production-ready programmer	Multi-operation CAM project
6 Months	500–600	Manufacturing engineer track	Portfolio of 8+ programs

2-Week Crash Course

Days	Topics	Deliverable
D1–2	CNC machine types, coordinates, safety	Safety checklist + setup sheet
D3–4	G-code, M-code, linear/circular interpolation	Manual G-code programs
D5–6	CAM: facing, pocketing, contouring	2.5D CAM toolpaths
D7–8	Tool selection, speeds/feeds, offsets	Optimized toolpath sheet
D9–10	Capstone	CNC program + simulation video

1-Month Foundation

Week	Topics	Project
W1	Lathe programming (turning, threading)	Lathe G-code portfolio
W2	Mill programming (pockets, drilling cycles)	Mill program set
W3	CAM software: toolpath optimization	Complex part CAM
W4	Setup sheets, QC inspection, capstone	Production-ready CNC package

3–6 Month Progression

3-axis advanced + 5-axis intro + fixture design + SPC/quality + shop floor documentation (6-month).

Assessment Rubric

Weekly Quiz 20% | Programming labs 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

Tier 3 **CNC Programming**. Simulation-first; physical machine optional for 6-month tier with shop partnership.